

Cloud Computing and Virtualization

Cloud Computing and Virtualization

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Lecture – 04

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In this lecture...

I. Virtualization

- I. Virtualization Technologies: Virtual Machines, Hypervisors(Type1, Type2)
- II. Overview of Containerization and Kubernetes

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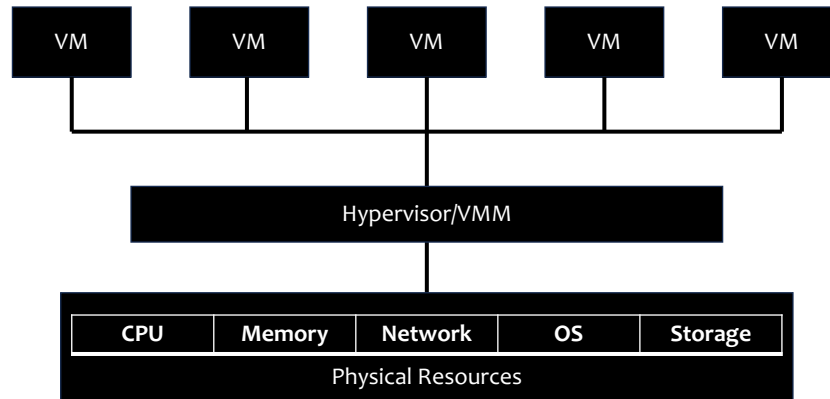
Virtualization

- Virtualization allows one physical machine to act like many separate machines.
- Cloud computing needs to support multiple customers, scale resources, and run different workloads at the same time.
- Virtualization makes this possible by dividing a physical machine's resources (CPU, RAM, storage) into isolated units.
- The technology that works behind virtualization is known as a virtual machine monitor(VMM) which separates compute environments from the actual physical infrastructure.
- Virtualization improves resource efficiency, ensures isolation between workloads, enables scalability by quickly adding or removing virtual machines, provides flexibility to run multiple operating systems on the same hardware, reduces costs through optimized hardware usage, and supports disaster recovery by allowing easy backup, migration, and replication of virtual machines.

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Virtualization



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Virtual Machine Monitor

- Virtual machine monitor (VMM) is a software program that enables the creation, management and governance of virtual machines (VM) and manages the operation of a virtualized environment on top of a physical host machine.
- Virtual Machine manager is also known as Hypervisor.
- A computer on which a hypervisor runs one or more virtual machines is called a host machine and each virtual machine is called a guest machine.
- Example: Oracle Virtual Box

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Virtual Machine Monitor – Hypervisor

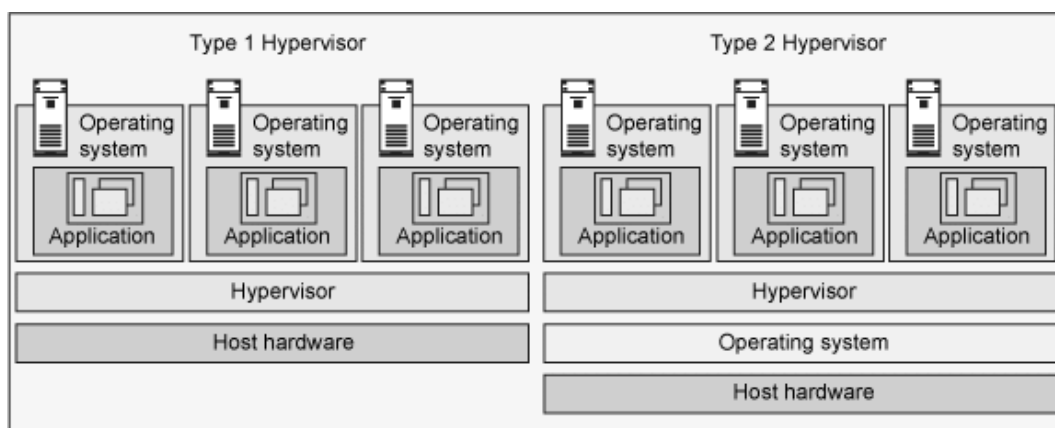
Hypervisor Types

- **Type 1 hypervisor or Bare metal hypervisor** is deployed directly over the host's system hardware without any underlying operating systems.
- Eg. Microsoft Hyper-V
- **Type 2 hypervisor or host based hypervisor** run on a host operating system that provides virtualization services such as I/O device support, memory management.
- Eg. Oracle Virtual Box

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Virtual Machine Monitor – Hypervisor



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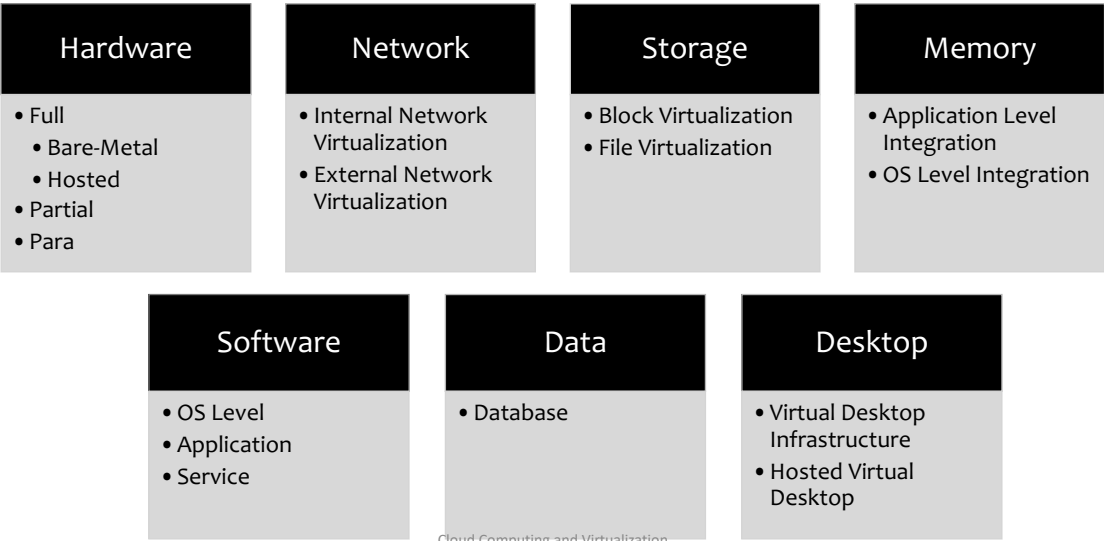
Virtual Machine Monitor – Hypervisor

Type-1 Hypervisor	Type-2 Hypervisor
Installed directly on physical hardware	Installed on top of an existing operating system
High performance, low latency	Lower performance due to OS overhead
Used in Data centers, cloud providers, enterprise servers	Used for personal use, testing, development
More secure (no host OS layer)	Less secure (depends on host OS security)
Highly scalable for large workloads	Suitable for small-scale environments
Very stable and reliable for production	Stability depends on host OS performance

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Types of Virtualization



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Hardware Virtualization

• Full Virtualization

- A hypervisor fully simulates the underlying hardware so that each virtual machine (VM) can run its own unmodified operating system.
- Allows maximum isolation and security.

• Partial Virtualization – Emulation Virtualization

- The virtual machine simulates the hardware and hence becomes independent of it.
- The guest operating system does not require modification.

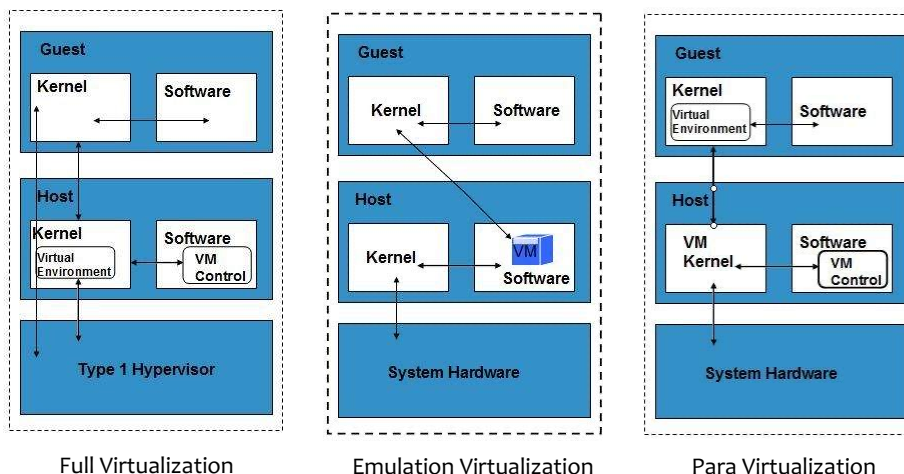
• Para – Virtualization

- The hardware is not simulated. The guest software run their own isolated domains.
- OS is modified to interact directly with the system hardware.

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Hardware Virtualization



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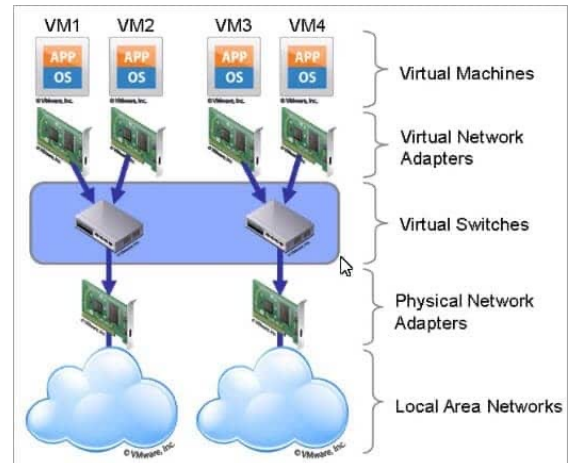
Network Virtualization

• Internal Network Virtualization

- Virtual networks created within a single physical server or data center.
- Includes VLANs, virtual switches, virtual routers
- Used for connecting VMs inside the same host or cluster
- Improves isolation, segmentation, and testing environments

• External Network Virtualization

- Combines multiple physical networks or servers into one virtualized network.
- Used in large cloud environments
- Enables creation of Virtual Private Clouds (VPCs), VPNs, SDN (Software Defined Networking)
- Provides secure multi-tenant networking



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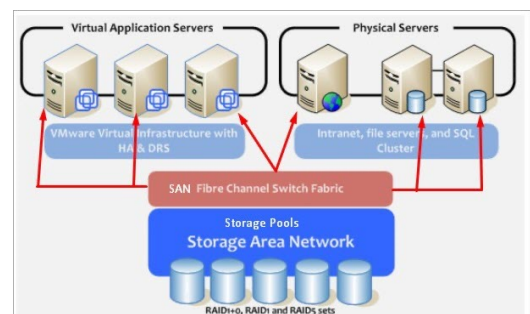
Storage Virtualization

• Block Virtualization

- Abstracts physical block devices (HDD/SSD) into logical storage units.
- Used in SAN (Storage Area Networks)
- Cloud block storage: Amazon EBS, Azure Managed Disks
- Provides redundancy, pooling, and better performance

• File Virtualization

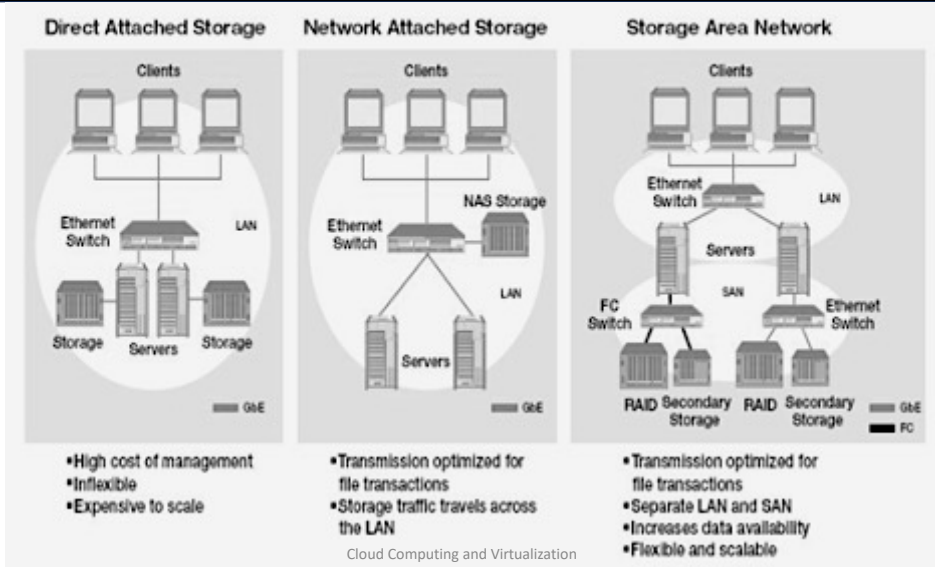
- Abstracts file-level storage across multiple file servers.
- Users see one unified file system
- Used in Network Attached Storage (NAS) systems
- Examples: Google File System (GFS)



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DAS vs NAS vs SAN

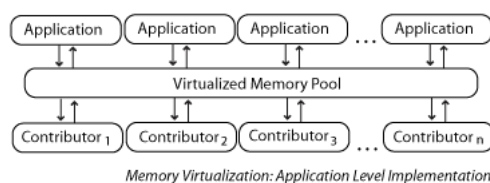


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Memory Virtualization

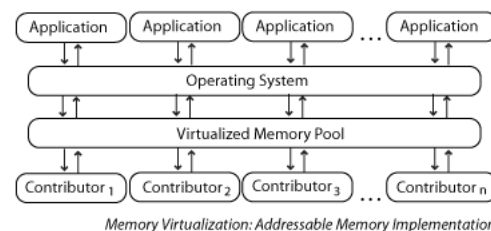
Application-Level Integration

- Applications utilize virtual memory mechanisms independently.
- Example: Java Virtual Machine (JVM) manages memory on its own
- Provides isolation for application processes



OS-Level Integration

- The operating system provides virtual memory using techniques like; Paging, Segmentation, Swapping, etc.



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Software Virtualization

- **OS-Level Virtualization**
 - Multiple OS environments run on the same kernel.
 - Containers (Docker, LXC)
 - Lightweight, fast, ideal for microservices
- **Application Virtualization**
 - Applications run in isolated environments without being installed on the host.
 - Prevents application conflicts
 - Useful for enterprise app deployment
- **Service Virtualization**
 - Simulates dependent services or APIs for testing.
 - Used in software testing when database, server, or API is unavailable
 - Example: mocking backend services in QA

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Data Virtualization

- **Database Virtualization**
 - Creates a virtual data layer that allows access to data from different sources as a single database.
 - No need to move or copy data
 - Used in big data and analytics
 - Tools: Denodo, IBM Infosphere
- **Benefits:**
 - Real-time data access
 - Eliminates duplication
 - Simplifies reporting and BI(Business Intelligence)

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Desktop Virtualization

- Virtual Desktop Infrastructure
 - Desktop operating systems are hosted on central servers; users access them remotely.
 - Example: VMware Horizon, Citrix VDI
 - Best for labs, companies, and remote work
 - Centralized control, secure, easy backup
- Hosted Virtual Desktop
 - A desktop OS runs on a cloud provider's infrastructure.
 - Example: Amazon WorkSpaces, Azure Virtual Desktop
 - Subscription-based virtual machines accessible from anywhere

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Assignment – 01

Choose any one existing Virtual Machine or Hypervisor (such as VMware ESXi, KVM, Xen, Hyper-V, VirtualBox, QEMU, or Proxmox) and prepare a detailed case study explaining its architecture, working mechanism, key features, advantages, limitations, and real-world use cases. Include diagrams, screenshots (if possible), and compare it with at least one alternative virtualization technology.

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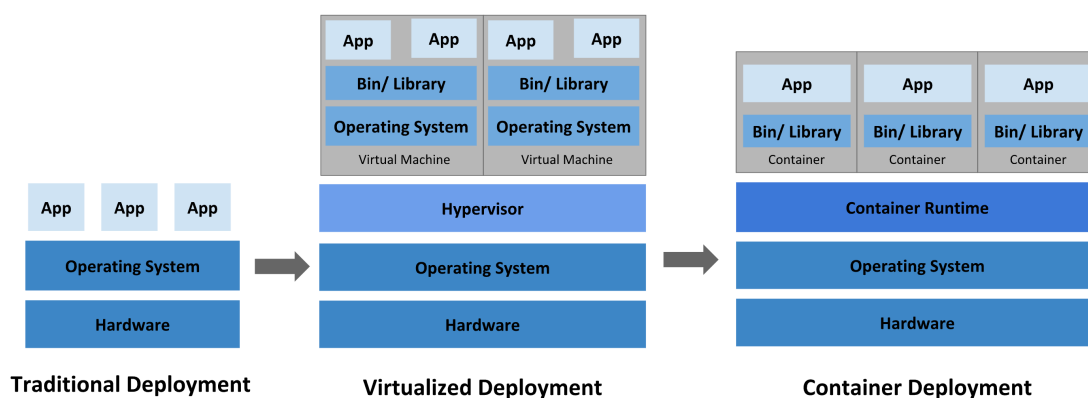
Containerization

- Containerization is a software deployment process that bundles an application's code with all the files and libraries it needs to run on any infrastructure.
- This self-contained package, or "container" is lightweight and portable because it doesn't need its own guest operating system. Instead, it shares the kernel of the host operating system while running in its own isolated user space.
- This isolation means you can run multiple containers on a single host, each with its own set of dependencies, without worrying about conflicts between them.

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Virtual Machine vs Containers



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Virtual Machine vs Containers

Virtual Machines	Containers
Virtualize hardware resources	Virtualize the operating system (OS)
Larger footprint, consume more resources	Lightweight, consume fewer resources
Strong isolation, each VM has its own OS	User space isolation, share OS kernel
Less portable, include full guest OS	Highly portable, encapsulate app and dependencies
Complex management, hypervisor-based management tools	Easier to manage, orchestration with tools like Kubernetes

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Benefits of Containerization

- **Portability:** Build once and run the containerized application across multiple environments and operating systems without rewriting code.
- **Scalability:** Containers are lightweight, start quickly, and allow running many isolated applications on the same machine efficiently.
- **Fault tolerance:** Isolated containers ensure that failure in one container doesn't affect others, improving application resilience and availability.
- **Agility:** Developers can update, troubleshoot, and release software faster because containers run independently from the underlying OS and hardware.

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Containerization Use Cases

Microservices
Architecture

Application
Modernization:

DevOps &
CI/CD Pipelines

Multi-Cloud &
Hybrid
Deployments

Scalable Web
Applications

Isolated
Development
Environments

Disaster
Recovery

Edge & IoT
Computing

Testing Across
Environments

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Container Orchestration

- Container orchestration is a software technology that allows the automatic management of containers.
- This is necessary for modern cloud application development because an application might contain thousands of microservices in their respective containers. The large number of containerized microservices makes it impossible for software developers to manage them manually.
- Developers use container orchestration tools to automatically start, stop, and manage containers.
- Container orchestrators allow developers to scale cloud applications precisely and avoid human errors.
- Kubernetes or K8s is one of the most popular container orchestration tools.

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Kubernetes

- Kubernetes, the most popular container orchestration tool available, is an open-source technology (originally open-sourced by Google, based on their internal project called Borg) that automates Linux container functions.
- Kubernetes works with many container engines, like Docker Engine.
- Docker is an open-source platform that enables developers to build, deploy, run, update and manage containers. Docker Engine is the runtime for building and running containers.

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Lecture – 05

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In this lecture...

- I. Virtual Machines (VM)
 - VM management and configuration
- II. Virtualization levels
 - Hardware, Software, Operating System, Data and Networking

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VM Management and Configuration

- VM Management is the process of creating, configuring, operating, monitoring, securing and maintaining virtual machines in cloud and on-premise environments.
- In cloud computing, VM management ensures resources (CPU, RAM, storage, network) are properly allocated and optimized throughout the VM lifecycle.

VM Provisioning

Provisioning means setting up and launching a new virtual machine.

- Select OS image (Linux/Windows)
- Choose hardware: vCPU, RAM, disk size
- Select network: VPC/VNet, subnet, firewall rules
- Use tools: Cloud console, CLI (aws cli, az cli), or automation tools (Terraform)
- Provisioning types: manual, automated (IaC), template-based

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VM Configuration

Compute Configuration

Configuring CPU and RAM for optimal performance:

- vCPU allocation: number of virtual CPU cores assigned
- Memory allocation: fixed or dynamic
- Resource control:
 - Limits – maximum resource a VM can use
 - Reservations – guaranteed minimum resource
 - Shares – priority assigned when multiple VMs compete

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VM Configuration

Storage Configuration

Managing the VM's virtual hard disks:

- Disk types:
 - Virtual disks (VMDK, QCOW2, VHD)
 - Cloud block storage (AWS EBS, Azure Disk)
- Provisioning methods:
 - Thin (grows as needed)
 - Thick (allocated fully at creation)
- Disk resizing, partitioning, formatting, mounting

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VM Configuration

Network Configuration

Setting up how the VM communicates:

- Assigning virtual NICs
- Connecting to virtual switches/bridges
- Setting static or dynamic IP
- Configuring SSH (Linux) / RDP (Windows) for remote access
- Applying firewall rules, security groups

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VM Configuration

VM Images and Snapshots

Images: Preconfigured templates used to create VMs.

- Examples: Ubuntu 22.04 image, Windows Server image.

Snapshots: A point-in-time backup of a VM's disk. Used to:

- Roll back after updates
- Clone VMs
- Create backups
- Snapshots are fast but not a full backup solution.

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VM Configuration

VM Security Management

Ensuring VM safety and protection:

- Access control: SSH keys, IAM roles, strong passwords
- Firewall rules: open only required ports
- Patching OS: regular updates
- Hardening: disable unused services, restrict root login
- Antivirus and malware protection
- Network isolation using security groups/VLANs

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VM Configuration

VM Migration

Moving a VM from one host to another.

Types:

- Cold migration: VM is stopped before moving
- Live migration: VM moves while running, no downtime
- Storage migration: Moving disks to another storage device

Reasons:

- Load balancing
- Hardware maintenance
- Disaster recovery
- Scaling infrastructure

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VM Configuration

VM Scaling & Availability

Ensuring applications stay available and can grow.

- Vertical Scaling
 - Add more CPU/RAM to existing VM
 - Example: upgrade from 2 vCPU to 8 vCPU
- Horizontal Scaling
 - Add more VMs to handle load
 - Example: multiple web server VMs behind a load balancer

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VM Configuration

VM Monitoring and Performance Tuning

Monitoring tracks VM resource usage:

- CPU usage
- Memory usage
- Disk I/O
- Network traffic

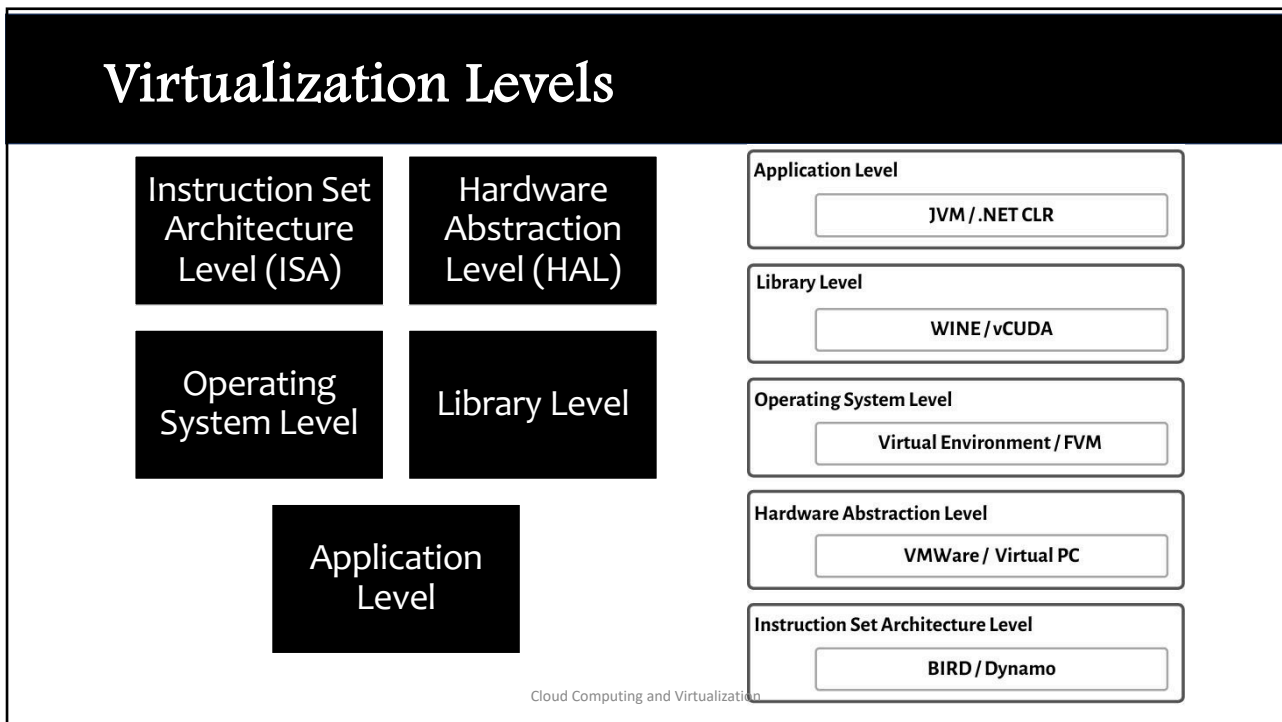
Tools: CloudWatch (AWS), Azure Monitor, Prometheus, htop/iostat.

Performance tuning involves:

- Adjusting CPU/RAM size
- Using faster storage (SSD/NVMe)
- Improving network drivers (VirtIO/SR-IOV)
- Removing unnecessary background tasks

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Instruction Set Architecture Level (ISA)

- In ISA, virtualization works through an ISA emulation.
- This is helpful to run loads of inherited code which was originally written for different hardware configurations.
- These codes can be run on the virtual machine through an ISA.
- The basic emulation, though, requires an interpreter. This interpreter interprets the source code and converts it into a hardware-readable format for processing.

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Hardware Abstraction Level (HAL)

- As the name suggests, this level helps perform virtualization at the hardware level. It uses a bare hypervisor for its functioning.
- This level helps form the virtual machine and manages the hardware through virtualization.
- It enables virtualization of each hardware component such as I/O devices, processors, memory, etc.
- This way multiple users can use the same hardware with numerous instances of virtualization at the same time.

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Operating System Level

- At the operating system level, the virtualization model creates an abstract layer between the applications and the OS.
- It is like an isolated container on the physical server and operating system that utilizes hardware and software. Each of these containers functions like servers.
- When the number of users is high, and no one is willing to share hardware, this level of virtualization comes in handy.

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Library Level

- OS system calls are lengthy and bulky, which is why applications opt for APIs from user-level libraries.
- Library-interfacing virtualization is made possible by API hooks. These API hooks control the communication link from the system to the applications.

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Application Level

- Application-level virtualization comes in handy when you only need to virtualize an application. It does not virtualize an entire platform or environment.
- On an operating system, applications work as one process. Hence it is also known as *process-level virtualization*.
- It is generally useful when running virtual machines with high-level languages. Here, the application sits on top of the virtualization layer, which is above the application program.

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Lecture – 06

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In this lecture...

- I. Virtual Private Cloud (VPC)
- II. Virtual Private Network (VPN)
- III. DNS Service and Cloud Network
- IV. Multihoming

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Virtual Private Cloud

- A virtual private cloud (VPC) is a secure, isolated private cloud hosted within a public cloud.
- VPC customers can run code, store data, host websites, and do anything else they could do in an ordinary private cloud, but the private cloud is hosted remotely by a public cloud provider.
- VPCs combine the scalability and convenience of public cloud computing with the data isolation of private cloud computing.
- Imagine a public cloud as a crowded restaurant, and a virtual private cloud as a reserved table in that crowded restaurant. Even though the restaurant is full of people, a table with a “Reserved” sign on it can only be accessed by the party who made the reservation. Similarly, a public cloud is crowded with various cloud customers accessing computing resources but a VPC reserves some of those resources for use by only one customer.

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Virtual Private Cloud Implementation

- VPC isolates computing resources from the other computing resources available in the public cloud. The key technologies for isolating a VPC from the rest of the public cloud are:
 - **Subnets:** A subnet is a range of IP addresses within a network that are reserved so that they're not available to everyone within the network, essentially dividing part of the network for private use. In a VPC these are private IP addresses that are not accessible via the public Internet.
 - **VLAN:** A VLAN is a virtual LAN. Like a subnet, a VLAN is a way of partitioning a network, but the partitioning takes place at a different layer within the OSI model (data link layer instead of network layer).
 - **VPN:** A virtual private network (VPN) uses encryption to create a private network over the top of a public network. VPN traffic passes through publicly shared Internet infrastructure routers, switches, etc. but the traffic is scrambled and not visible to anyone.

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Advantages of VPC over Private Cloud

- **Scalability:** Because a VPC is hosted by a public cloud provider, customers can add more computing resources on demand.
- **Easy hybrid cloud deployment:** It's relatively simple to connect a VPC to a public cloud or to on-premises infrastructure via the VPN.
- **Better performance:** Cloud-hosted websites and applications typically perform better than those hosted on local on-premises servers.
- **Better security:** The public cloud providers that offer VPCs often have more resources for updating and maintaining the infrastructure, especially for small and mid-market businesses.

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Virtual Private Network

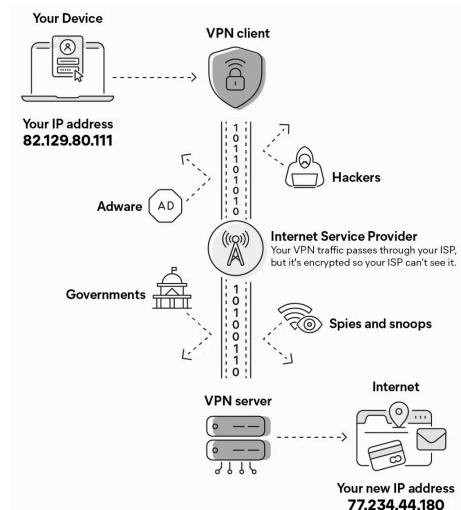
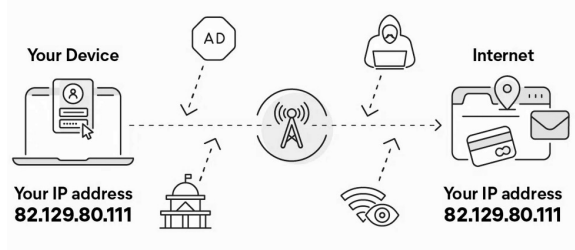
- A virtual private network, or VPN, is an encrypted connection over the Internet from a device to a network. The encrypted connection helps ensure that sensitive data is safely transmitted.
- It allows users to access remote networks safely and protect their online data from hackers, ISPs, or public Wi-Fi risks.
- VPN works through three main steps:
 - Encryption of Data: When you connect to a VPN, it encrypts your internet traffic.
 - VPN Tunnel Creation: Your device connects to a VPN server through a secure tunnel.
 - Routing Through VPN Server: Your internet traffic goes through the VPN server, not directly from your device.

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Virtual Private Network

- VPN servers does:
 - Assigns you a new IP address
 - Masks your real IP
 - Forwards your requests to websites
 - Sends the information back to you securely



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Virtual Private Network – Advantages

- Privacy & Anonymity
 - Hides your real IP address
 - Prevents tracking by websites, ISPs, or advertisers
- Data Security
 - Encrypts your connection
 - Protects you on public Wi-Fi (cafes, airports)
- Remote Access
 - Employees can securely access company internal networks
- Access Restricted Content
 - Bypass geo-blocking
 - Access content available only in certain countries
- Improved Online Safety
 - Ideal for banking, payments, or confidential communication

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Virtual Private Network – Disadvantages

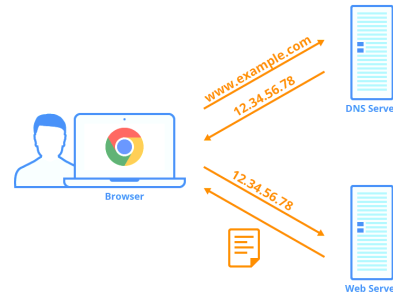
- Potentially slower speeds
- QoS (quality of service) challenges
- No cookie protection: VPN can't block cookies
- Not total privacy: The VPN provider itself can see your internet activity if they want.

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DNS Service

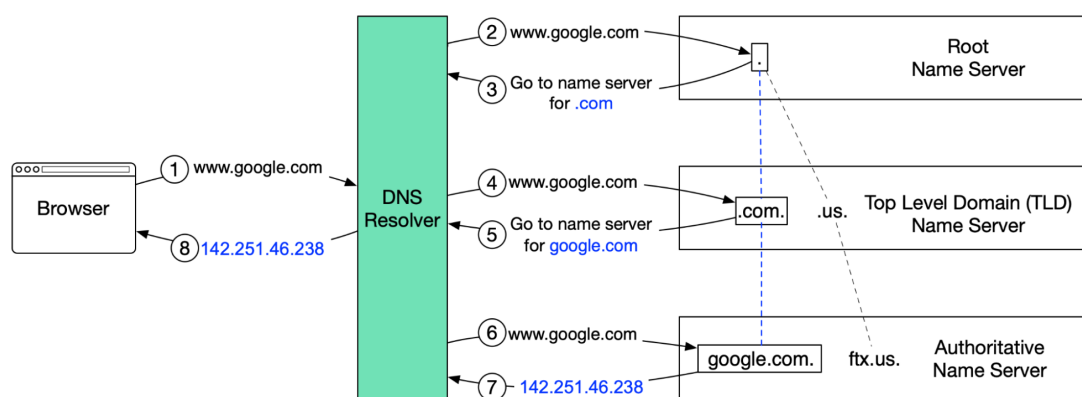
- The Domain Name System (DNS) is the phonebook of the Internet.
- When users type domain names into the URL bar in their browser, DNS servers are responsible for translating those domain names to numeric IP addresses, leading them to the correct website.



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How DNS resolve IP?



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DNS Service

- Client (stub resolver) asks its configured recursive resolver (e.g., ISP DNS or 8.8.8.8): “What is A www.example.com?”
- If recursive resolver has the answer cached and TTL not expired return it.
- If not cached, the resolver performs the resolution:
 - Query a root server: “Who is responsible for .com?” Root replies with NS records for .com TLD servers.
 - Query a .com TLD server: “Who is authoritative for example.com?” TLD returns NS records for example.com authoritative servers.
 - Query one of the authoritative name servers for example.com: “What is A www.example.com?” Authoritative server replies with the A record (or an NXDOMAIN if it doesn’t exist).
- Recursive resolver returns the A record to the client and stores it in cache for TTL seconds.
- The client then connects to the returned IP.

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DNS Record Types

- DNS records, also known as zone files, store critical information about domains. These text-based files use the DNS syntax and are housed on DNS servers.
- DNS servers use the information in DNS records to route internet traffic.
- By configuring these records, we can optimize performance, enhance security, and tailor functionality to the specific needs of websites, applications, and networks.

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Common DNS Record Types

Record Types	Description
A	Maps a domain to an IPv4 address.
AAAA	Maps a domain to an IPv6 address.
CNAME	Aliases one domain name to another.
MX	Directs email to mail servers.
NS	Specifies the authoritative nameserver for a domain.
SOA	Contains administrative information about the domain, like the primary nameserver and zone update settings.
TXT	Stores text information, often used for verification.
SRV	Specifies a service location for certain services, like servers handling VoIP.
PTR	Maps an IP address to a domain name for reverse DNS lookups.

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Common DNS Record Types

A (Address) records

- They translate domain names into IP addresses and store these associations. A records can only hold IPv4 addresses.

Domain Name	Record Type	Value	TTL
nec.edu.np	A	192.0.0.1	14400

- **Domain name.** The website's domain name. In a DNS zone file, the @ symbol is often used instead of the full domain name to represent the root domain.
- **Record type.** Specifies the record type as A.
- **Value.** Contains the IP address associated with the domain name.
- **TTL.** Lists the record's Time to Live in seconds. The default value is 14400 seconds (240 minutes), which determines how long DNS resolvers cache the record.

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Common DNS Record Types

AAAA records

- It function like A records, but instead of storing IPv4 addresses, they exclusively store IPv6 addresses connected to domain names.

CNAME (Canonical Name) record

It serves as a replacement for an A record if a domain is an alias for another domain. Because of this, all CNAME records point to a domain name instead of an IP address.

Domain Name	Record Type	Value	TTL
nec.com.np	CNAME	nec.edu.np	14400

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Common DNS Record Types

MX (Mail Exchange) records

- It store instructions for directing emails to mail servers using the SMTP protocol.

Domain Name	Record Type	Priority	Value	TTL
gmail.com	MX	10	mail.google.com	14400

Priority: Specifies the preference for delivering mail, with lower values indicating higher priority. If the highest-priority mail server fails, the server redirects the mail to the next highest-priority server.

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Common DNS Record Types

NS (Nameserver) record

- It specifies which server is authoritative for the DNS records of a given domain. Domains should have multiple NS records pointing to different nameservers to provide redundancy and ensure high availability.

Domain Name	Record Type	Value	TTL
pu.edu.np	NS	NS1.NEPLINK.NET	14400

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Common DNS Record Types

SOA (Start of Authority)

- This record contains critical information about a DNS zone which play a key role in zone transfers between DNS servers.
- SOA records define the following zone properties:
 - **MNAME.** The primary nameserver for the zone.
 - **RNAME.** The email address of the person responsible for the zone.
 - **Serial.** The zone's serial number serves to track changes and synchronize zone data across DNS servers.
 - **Refresh.** The interval (in seconds) at which secondary DNS servers check the primary server for updates.
 - **Retry.** The time (in seconds) a secondary server waits before retrying a failed update from the primary server.
 - **Expire.** The time (in seconds) a secondary server will continue attempting to update from the primary server before considering the data stale and stopping updates.
 - **TTL.** This value determines how long DNS resolvers cache a DNS record before refreshing it.

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Common DNS Record Types

TXT (Text) records

- It store arbitrary text data. They are used for domain verification, sender policy framework (SPF), DKIM, or other forms of authentication.

Domain Name	Record Type	Value	TTL
pu.edu.np	TXT	Example Text	14400

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Common DNS Record Types

SRV (Service) records

- It stores essential information about host and port details for specific internet services, such as email, VoIP, and other applications. Certain protocols, like LDAP, SIP, and XMPP, require valid SRV records to function properly.

SRV records hold the following information:

- **Service.** The symbolic name of the service.
- **Protocol.** Specifies if the service is using TCP or UDP protocols.
- **Name.** The domain name associated with the service.
- **TTL.** The number of seconds DNS resolvers cache a DNS record before refreshing it.
- **Class.** Can contain IN (default), CH (used for querying DNS server versions), or HS (uses DNS functionality to provide access to databases).
- **Priority.** Determines the preferred server. Lower values indicate higher priority.
- **Weight.** Used when multiple servers have the same priority; higher values give a server more priority.
- **Port.** The TCP or UDP port the service is running on.
- **Target.** The canonical hostname of the machine providing the service.

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Common DNS Record Types

PTR (Pointer) records

- These are the inverse of A or AAAA records. They map IP addresses back to domain names and facilitate reverse DNS lookups.
- PTR records store IP addresses in reverse:
 - IPv4 addresses are saved with their octets in reverse order.
 - IPv6 addresses are saved in reverse order of their nibbles (each 4-bit section of the hexadecimal address)

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Other DNS Record Types

Record Type	Description
AFSDB	Specifies the location of Andrew File System (AFS) cells.
ATMA	Maps a domain name to an ATM address used for ATM networks.
CAA	Specifies which certificate authorities (CAs) are allowed to issue certificates for a domain.
CERT	Stores certificates and certificate-related information, such as public keys.
DHCID	Used in DHCP to associate DNS names with dynamically assigned IP addresses.
DNAME	Provides redirection of a subtree of the DNS namespace to another domain.
DNSKEY	Contains public keys used to verify DNSSEC signatures.
DS	Used in DNSSEC to identify a DNSKEY record in the delegated zone.

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Other DNS Record Types

Record Type	Description
HINFO	Provides information about the hardware and operating system used by a host.
ISDN	Stores ISDN addresses associated with a domain name.
MB, MG, MINFO, MR	Legacy records related to mailbox information, with specific uses for mapping and informational purposes.
NAPTR	Used for URI and ENUM applications to define rules for rewriting domain names.
NSAP	Maps a domain name to an NSAP address used in OSI networks.
NSEC	Used in DNSSEC to prove non-existence of a DNS record by listing the next record in the zone.
NSEC3	Enhanced version of NSEC for DNSSEC with hashed domain names to prevent enumeration.

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Other DNS Record Types

Record Type	Description
NSEC3PARAM	Stores parameters for NSEC3 records, including hashing algorithms and iterations.
RP	Provides information about the person responsible for a domain, including contact details.
RRSIG	Contains a cryptographic signature used to verify DNSSEC-signed data.
RT	Specifies a route through an intermediate host, used for non-IP networks.
TLSA	Links a domain name with a TLS certificate for DANE.
X25	Stores an X.25 network address used in older packet-switched networks.

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Multihoming

- Multihoming is connecting a device or network to multiple other networks (like different Internet Service Providers or internal networks) to boost reliability, performance, and fault tolerance by providing backup paths for data, preventing single points of failure, and ensuring continuous connectivity for critical applications.
- How it works:
 - **Multiple Connections:** A network has more than one link to the outside world, either to one ISP or multiple ISPs.
 - **Redundant Pathways:** If one link fails, traffic automatically reroutes through another, ensuring service continuity.
 - **Dynamic Protocols:** Uses protocols (like BGP – Broader Gateway Protocol) to manage routes and switch traffic between connections.

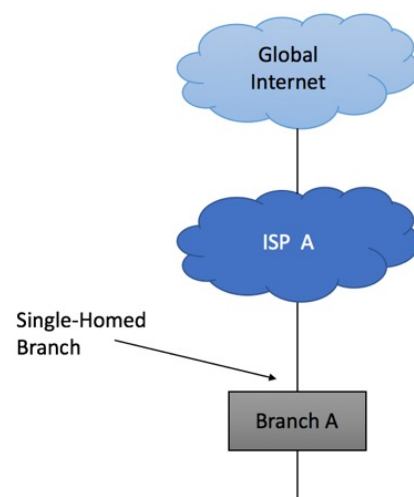
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Multihoming

Single-homed Network :

- In a single-homed network you have only one external link (the usual home or small-office setup). That's simple and cheap.
- But if that link goes down — e.g. due to fiber cut, ISP outage, maintenance or DDoS — the entire site loses Internet access.
- For businesses needing high availability, relying on a single link is a recipe for disaster.

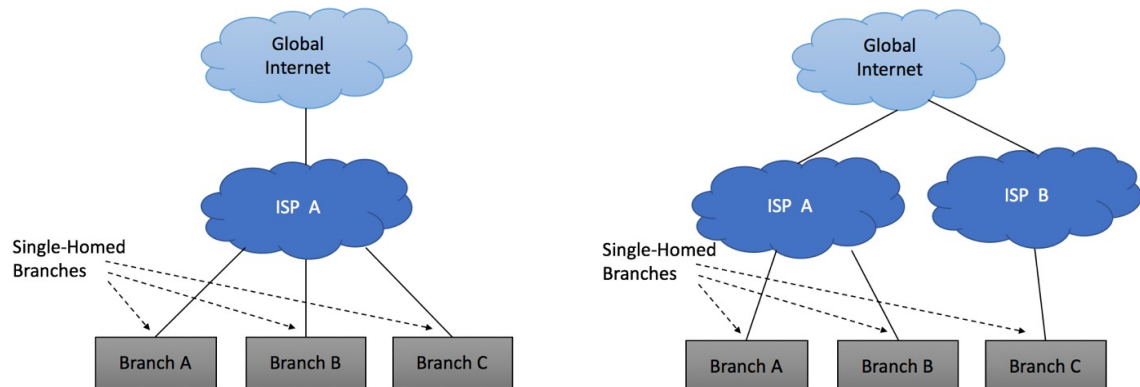


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Multihoming

Single-homed Network :



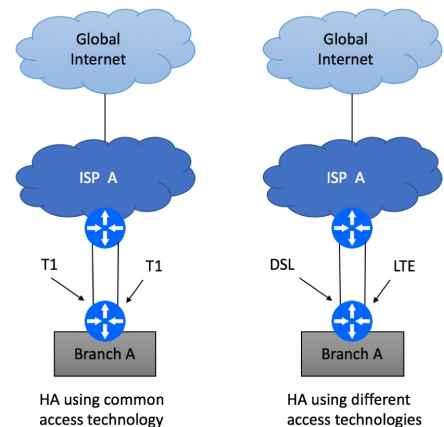
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Multihoming Implementation Scenarios

Multi-Homing Scenario 1 – Same PE / CE

- Two physical links (e.g. two circuits) but both terminate on the same devices at both ends.
- This gives minimal redundancy: if those devices fail, connection still drops.
- Might be used where limited options exist.



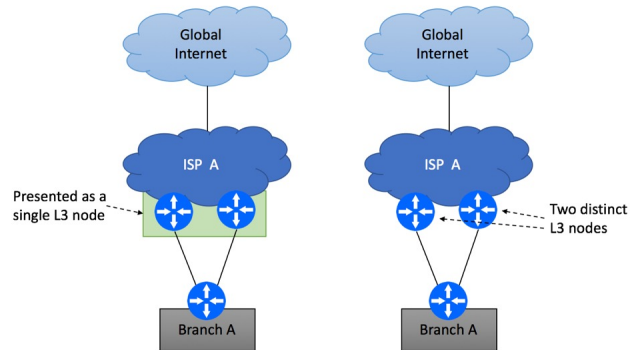
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Multihoming Implementation Scenarios

Scenario 2 – Different PEs / Single CE

- The ISP side uses different PE (Provider Edge) devices; customer site uses a single Customer Edge (CE) router.
- Improves ISP-side resilience if one PE fails, the other handles traffic.
- Static routing or BGP can be used.



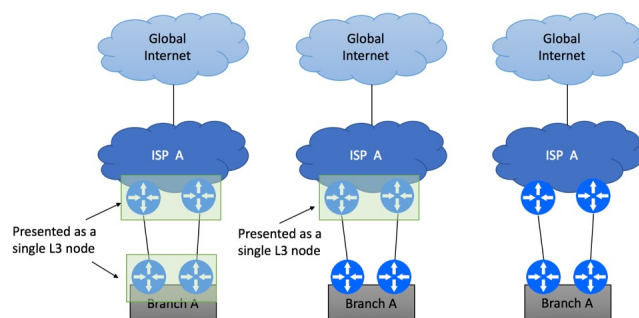
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Multihoming Implementation Scenarios

Scenario 3 - Different PEs and CEs

- Redundancy on both ends: multiple ISP devices and multiple routers on customer side.
- More complex, but offers better reliability, failover, high availability.



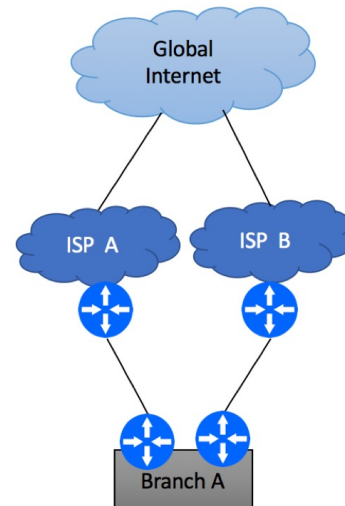
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Multihoming Implementation Scenarios

Scenario 4 - Multi-ISP (Different Providers)

- The strongest form of multihoming: connect to different ISPs.
- This guards against provider-specific failures (outages, policy changes, network problems).
- Typically uses routing protocols (e.g. Border Gateway Protocol BGP) to manage advertisement of IP prefixes, failover, routing decisions.



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When You Need BGP Multihoming

- if you:
 - Host **Internet-facing services** (web servers, mail servers, VPN gateways, etc.)
 - Want **public IPs that remain reachable** even if one ISP/connection fails
 - Need **real load balancing or stable routing across providers**then you likely need proper multihoming using BGP.
- That involves having your own IP prefix (or at least getting permission from your ISP to advertise your IP block to the global Internet)
- Getting an Autonomous System Number (AS) or using an AS provided by a registry or your ISP (depending on policy) so you can run BGP with multiple upstream providers.
- Without that, you risk losing reachability when links fail, or having services go down.

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END OF UNIT – 02

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